LUMAFLOW · USER REQUIREMENTS

LumaFlow user requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Kit 1.8 upgrade: user requirements drawn from the users, stakeholders and constraints in LMF-PRB-001 v0.9

LumaFlow user requirements

What users need, written from their side. Product requirements that meet these needs are in PRS (`03-requirements.md` , LMF-PRS-001) and the links are in the traceability matrix (`03b-traceability.md` , LMF-RTM-001). Method: STANDARDS section 22. The sources are the user and stakeholder tables of LMF-PRB-001 (Tables 1 and 2); these needs are proposals until they are reviewed with the co-design partner, a university extension programme for private well owners (LMF-DEC-001), which has not yet been approached.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Performance	Safer water at one tap	Water from my drinking tap has bacteria and protozoan cysts reduced to a level I can trust, in the clear, filtered well water I have	High	Well household (PRB Table 1)
UR-002	Performance	Useful flow	The tap fills a 1 L bottle in under a minute	High	Well household; clinic (PRB Table 1)
UR-003	Performance	Poorer water	If my water is cloudier than the design water, the unit still treats it or tells me that it cannot	Medium	Municipal household with events (PRB Table 1)
UR-004	Safety	No silent failure	The unit never passes untreated water without stopping it and telling me	High	All households; clinic (PRB Table 1)
UR-005	Usability	Instant on	Water is treated the moment I open the tap, with no warm-up and nothing to switch on	High	Well household (PRB Table 1)
UR-006	Environment and durability	No mercury	Nothing in the unit can release mercury into my water or home	High	Well household (PRB problem statement)
UR-007	Environment and durability	Household pressure	The unit copes with the pressure in my pipes, including surges, without leaking or breaking	High	Well household; installer (PRB Tables 1 and 2)
UR-008	Usability	Tap pressure	The unit does not noticeably weaken the flow at my tap	Medium	Well household (PRB Table 1)
UR-009	Cost	Low running energy	It uses little electricity, and almost none when the tap is closed	Medium	Well household; off-grid users (PRB Table 1)
UR-010	Environment and durability	Long life	It keeps working for years without parts wearing out from heat	Medium	Clinic with no technician (PRB Table 1)
UR-011	Safety	Nothing added	Nothing in the unit adds harmful substances to the water	High	All users (PRB constraints)
UR-012	Safety	No UV exposure	No one can be exposed to the UV-C light, even when opening the unit	High	All users; maker (PRB Table 1)
UR-013	Safety	Electrical safety	It is safe to have in a wet cabinet next to water pipes	High	All users (PRB constraints)
UR-014	Deployment and assembly	Fits under the sink	It fits under a kitchen sink beside the cold line	High	Well household; installer (PRB Tables 1 and 2)
UR-015	Maintenance	Easy service	I or a local helper can replace the parts that wear with hand tools, quickly	Medium	Clinic; maker (PRB Table 1)
UR-016	Cost	Affordable and buildable	A maker can build it from catalogue parts at a cost a household could consider	Medium	Maker; household (PRB constraints)
UR-017	Usability	Status at a glance	I can see at a glance that it is working and when it needs service	Medium	Clinic; household (PRB Table 1)

ID	Category	Title	Need	Priority	Source
UR-018	Regulatory	Honest claims	The unit and its documents say plainly what it is, a research and educational prototype, and never claim certified safe water	High	Certification bodies; all users (PRB Table 2 and constraints)

Categories: deployment and assembly, performance, usability, safety, environment and durability, logistics, maintenance, cost, regulatory. Priorities: High, Medium, Low.